Week - III ⇒ OS CONCEPTS..

* Threads vs Processes
* CPU scheduling & context switching
* I/O Blocking & Async I/O
* Virtual Memory & Paging
* Filesystem & Storage..

* Threads vs Process *

Firstly program.

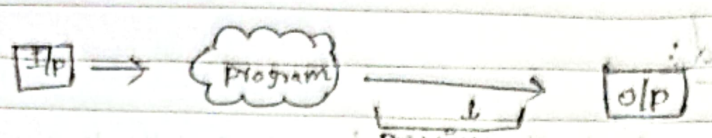
A program is a sequence of instructions written in a Programming language this code will been interpreted or compiled into binary code
↓
01 loaded to memory & execute the running

* Program can be a simple script code or an group of program will called as Software..
↓
IN which has blocks of programs

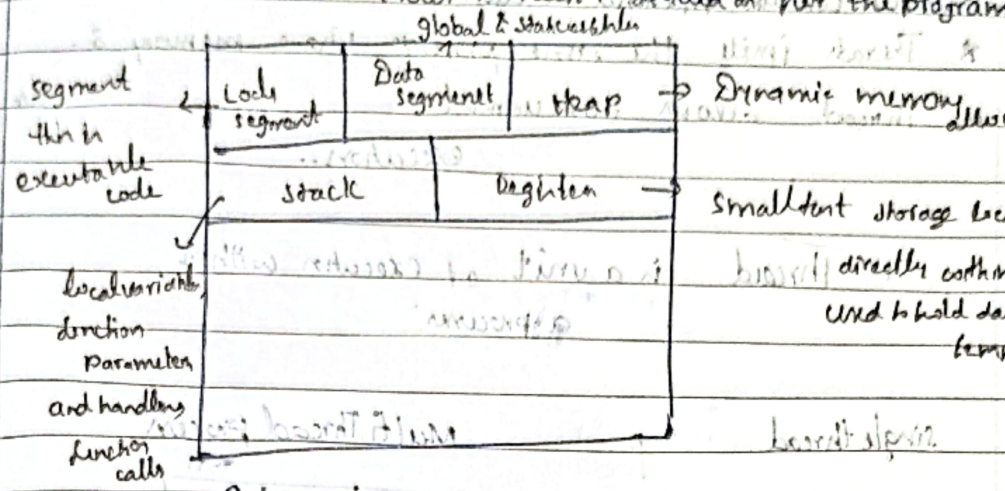
Process :-

A process is simply an instance of a program that is being executed

* Program under execution / currently running state is called as Process



A Program will be under execution is called as process &
Process is held in divided as per the program



Each Process has this thing.

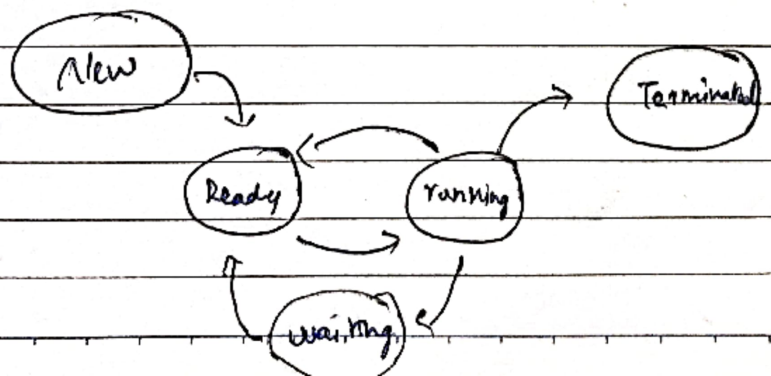
☆☆ Seperate memory space

→ one process cannot corrupt another process..

Ex
- chrome
- vscode
- Spotify

} → Each running program in a seperate process

A process has it own:
- memory space
- resources
- cpu state



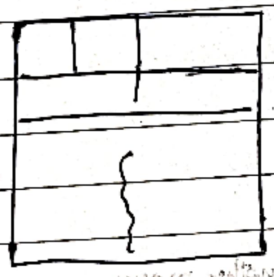
Thread:-

A thread is the smallest unit of execution inside a process.

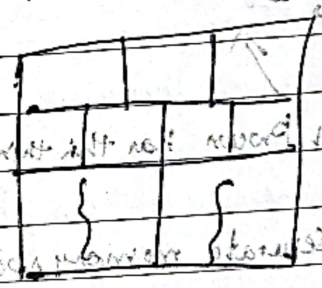
* Process contain multiple threads
* Theads inside the same process share memory & resources
* Thread allow concurrent execution..

Thread is a unit of execution within a process

single thread



Multi Thread process



* Each thread has their own Reg & stack but the code segment data segment, heap are same where they share resources..

* Shared memory space

* Efficient communication between threads

* ★ X A single thread can corrupt the entire process.

Ex

when watching Youtube

+ one thread loads video
+ one thread handles UI
+ one thread download data.

without thread the application would freeze..

Process	Thread
+ Independent program execution	+ Part of process
+ own memory	+ Shared memory
+ Heavy weight	+ light weight
+ slower execution	+ faster execution
+ seperate memory	+ Shared memory
+ Ipc needed for communication	+ Inter communication